

ANSWER KEY — Integration Lesson 34 Test: Volumes I — Disks & Washers

For the grader · full working shown · give credit for a correct setup with small arithmetic slips · setup-only questions earn full credit with no evaluation

1. Region under $y = x^2$ on $[0, 2]$ revolved about the x-axis. Volume?

Work: disk radius $r(x) = x^2$. Square FIRST: $(x^2)^2 = x^4$ (exponents multiply — not x^2). $V = \pi \int_0^2 x^4 dx = \pi[x^5/5]_0^2 = \pi(32/5 - 0)$. **Answer: $V = 32\pi/5$.** **Grader note:** squaring after integrating $(\pi(8/3)^2)$ is the classic error — no credit for that route.

2. Region under $y = \sqrt{2x}$ on $[0, 3]$ revolved about the x-axis. Volume?

Work: radius $r(x) = \sqrt{2x}$. Squaring kills the root: $(\sqrt{2x})^2 = 2x$. $V = \pi \int_0^3 2x dx = \pi[x^2]_0^3 = \pi(9 - 0)$. **Answer: $V = 9\pi$.** **Grader note:** $(\sqrt{2x})^2 = 2x$, not $2x^2$ and not $4x$.

3. $y = 5$ on $[2, 6]$ revolved about the x-axis; check against $\pi r^2 h$.

Work: every disk is identical, radius 5. $V = \pi \int_2^6 5^2 dx = \pi \int_2^6 25 dx = 25\pi(6 - 2) = 100\pi$. Geometry: cylinder of radius $r = 5$ and height (length along the axis) $h = 6 - 2 = 4 \rightarrow \pi r^2 h = \pi \cdot 25 \cdot 4 = 100\pi$ ✓. **Answer: $V = 100\pi$, with $r = 5$ and $h = 4$.** **Grader note:** using $h = 6$ (forgetting to subtract the lower limit) gives 150π — wrong.

4. $y = x$ on $[0, 4]$ revolved about the x-axis; check against $(1/3)\pi r^2 h$.

Work: radius $r(x) = x$, so $(r)^2 = x^2$. $V = \pi \int_0^4 x^2 dx = \pi[x^3/3]_0^4 = \pi(64/3)$. Geometry: a cone lying on its side, tip at the origin, height $h = 4$ along the axis, base radius = height of the line at $x = 4$, so $r = 4$: $(1/3)\pi r^2 h = (1/3)\pi \cdot 16 \cdot 4 = 64\pi/3$ ✓. **Answer: $V = 64\pi/3$.** **Grader note:** the $1/3$ in the cone formula IS the 3 from $x^3/3$ — say so for full credit on the check.

5. Slab solid on $[0, 6]$ cm with $A(x) = 4 + x$ cm². (a) Volume? (b) Why cubic centimeters?

Work: (a) $V = \int_0^6 (4 + x) dx = [4x + x^2/2]_0^6 = (24 + 18) - 0 = 42$. (b) Each slab contributes (cross-section area) $\times$ (thickness) = cm² $\times$ cm = cm³, and adding slabs never changes the units. **Answer: (a) $V = 42$ cubic centimeters. (b) $A(x) dx$ carries cm² $\cdot$ cm = cm³.** **Grader note:** no π anywhere here — the slices are not stated to be circles, so $A(x)$ is used as given.

6. Region between $y = x + 1$ (top) and $y = 1$ (bottom) on $[0, 2]$, revolved about the x-axis. Volume?

Work: the floor of the region is $y = 1$, not the axis $\rightarrow$ washers. Outer $R = x + 1$, inner $r = 1$. Square each separately: $R^2 = (x + 1)^2 = x^2 + 2x + 1$ (keep the middle term!), $r^2 = 1$. $R^2 - r^2 = x^2 + 2x$. $V = \pi \int_0^2 (x^2 + 2x) dx = \pi [x^3/3 + x^2]_0^2 = \pi(8/3 + 4) = \pi(8/3 + 12/3) = 20\pi/3$. **Answer: $V = 20\pi/3$.**

Check: full solid $\pi \int_0^2 (x + 1)^2 dx = \pi [(x+1)^3/3]_0^2 = \pi(27/3 - 1/3) = 26\pi/3$, minus the drilled cylinder of radius 1 and length 2 ($= 2\pi = 6\pi/3$) $\rightarrow 26\pi/3 - 6\pi/3 = 20\pi/3$ ✓. **Grader note:** dropping the $2x$ cross term gives $14\pi/3$ — wrong.

7. Region between $y = 3x$ and $y = x$ on $[0, 1]$ revolved about the x-axis. Student writes $V = \pi \int_0^1 (3x - x)^2 dx$. (a) Student's value? (b) Correct value? (c) Why $R^2 - r^2 \neq (R - r)^2$?

Work: (a) Student: $(3x - x)^2 = (2x)^2 = 4x^2$, so $\pi \int_0^1 4x^2 dx = 4\pi [x^3/3]_0^1 = 4\pi/3$. (b) Correct washer: $R = 3x$, $r = x$, so $R^2 - r^2 = 9x^2 - x^2 = 8x^2$. $V = \pi \int_0^1 8x^2 dx = 8\pi [x^3/3]_0^1 = 8\pi/3$. (c) With $R = 3$, $r = 1$: $R^2 - r^2 = 9 - 1 = 8$, but $(R - r)^2 = 2^2 = 4$. Algebra agrees: $(R - r)^2 = R^2 - 2Rr + r^2$, which is never the same as $R^2 - r^2$ unless $r = 0$. **Answer: (a) $4\pi/3$ (b) $8\pi/3$ (c) $8 \neq 4$ — squaring the gap loses exactly half the volume here. Grader note:** a ring's area is the big circle MINUS the hole, $\pi(R^2 - r^2)$; $\pi(R - r)^2$ is the area of a small circle whose radius is only the gap width.

8. Region enclosed by $y = \sqrt{x}$ and $y = x^2$ revolved about the x-axis. Find the limits, then the volume.

Work: intersections: $\sqrt{x} = x^2 \rightarrow$ square both sides: $x = x^4 \rightarrow x^4 - x = 0 \rightarrow x(x^3 - 1) = 0 \rightarrow x = 0$ and $x = 1$. Which is outer? Test $x = 1/4$: $\sqrt{x} = 1/2$ versus $x^2 = 1/16$, so **$\sqrt{x}$ is the outer curve** ($R = \sqrt{x}$, $r = x^2$). Squares: $R^2 = x$, $r^2 = (x^2)^2 = x^4$. $V = \pi \int_0^1 (x - x^4) dx = \pi [x^2/2 - x^5/5]_0^1 = \pi(1/2 - 1/5) = \pi(5/10 - 2/10)$. **Answer: intersections $x = 0$ and $x = 1$; $V = 3\pi/10$. Grader note:** reversing R and r gives $-3\pi/10$ — a negative volume is an instant signal the curves were swapped.

9. Set up (do NOT evaluate) the volume when the region under $y = \sin x$ on $[0, \pi]$ is revolved about the x-axis.

Work: the region runs from the curve down to the x-axis $\rightarrow$ **disk method**. Radius $r(x) = \sin x$. Square it: $(\sin x)^2 = \sin^2 x$. Limits are given: 0 to π . **Answer: $V = \pi \int_0^\pi \sin^2 x dx$ — disk method, radius $r = \sin x$. Stop here; no evaluation required. Grader note:** $\sin^2 x$ means $(\sin x)^2$, NOT $\sin(x^2)$. Evaluating needs a power-reduction identity, so a correct setup earns the full point.

10. Region between $y = \sqrt{25 - x^2}$ and $y = 3$ revolved about the x-axis (hole drilled through a sphere). (a) Limits? (b) Volume?

Work: (a) $\sqrt{25 - x^2} = 3 \rightarrow$ square both sides: $25 - x^2 = 9 \rightarrow x^2 = 16 \rightarrow x = -4$ and $x = 4$. (b) Washers with $R = \sqrt{25 - x^2}$ (outer) and $r = 3$ (inner). Square first: $R^2 = 25 - x^2$, $r^2 = 9$, so $R^2 - r^2 = 16 - x^2$. $V = \pi \int_{-4}^4 (16 - x^2) dx = 2\pi \int_0^4 (16 - x^2) dx$ (even integrand, symmetry shortcut) $= 2\pi[16x - x^3/3]_0^4 = 2\pi(64 - 64/3) = 2\pi(192/3 - 64/3) = 2\pi(128/3)$. **Answer: (a) $x = \pm 4$ (b) $V = 256\pi/3$.** **Check:** less than the whole sphere, $(4/3)\pi \cdot 125 = 500\pi/3$ ✓ (the drilled-out part accounts for the rest). **Grader note:** a very common slip is integrating from 0 to 4 only, giving $128\pi/3$ — exactly half the ring. Also $(\sqrt{25 - x^2} - 3)^2$ is the forbidden squared-difference.